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Original article

Changes in the pattern of breast cancer burden among African American women: evidence based on 29 states and District of Columbia during 1998 to 2010

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A R T I C L E I N F O

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Purpose: Assessment of breast cancer (BC) pattern in individual states with respect to ethnicity.
Methods: Population-based cancer registries from the Cancer Incidence in Five Continents databases (1998–2007) supplemented with Surveillance, Epidemiology, and End Results data from 2008 to 2010 were used.
Results: The age-specific burden showed a clear convergence of BC burden among African American (AA) and Caucasian American (CA) in most states. This was primarily because of a decrease in the BC rate among CA aged 50 years or older and an increase among AA of the same age group. The 2003–2007/1998–2002 rate ratio for CA was 0.91 (95% confidence interval [CI], 0.90–0.91) in the South, whereas it was 1.06 (95% CI, 1.04–1.08) for AA. This convergence was confirmed in states with available data for the period 2008 to 2010. The AA/CA rate ratio among women aged younger than 40 years was 0.99 (95% CI, 0.99–1.04) in the Northeast, 1.29 (95% CI, 1.25–1.33) in the South, and 1.10 (95% CI, 1.04–1.17) in the West. This pattern correlates with the estrogen receptor positive and progesterone receptor positive pattern. The strongest disparity in estrogen receptor negative was observed in Louisiana which with Detroit, have had the highest rates of estrogen receptor negative.
Conclusions: The changes in postmenopausal hormone use and mammography screening might have played a role in the observed convergence.

Introduction

African-American women (AA) experience an overall lower breast cancer (BC) burden compared with that of Caucasian-American women (CA), but have the highest mortality burden [1,2]. It has been suggested that this higher mortality is partially due to a lower socioeconomic status, which prevents AA from routine mammography screening [3–5]. However, a recent study revealed that mammography screening among AA is higher than among CA, with an average of 64% for AA aged 40 years or older versus 59% for CA [1]. A study published in 2006 assessing BC trend in the United States by ethnicity reported that increases in BC incidence rate were limited to CA aged 50 years or older, and that the incidence rate for AA of the same age range was stable [6]. A different study analyzing data from 2006 to 2010 from different databases (including Surveillance, Epidemiology, and End Results (SEER) Program, National Cancer Registry, and North American Association of Central Cancer Registries) reports that BC incidence rates among AA and CA aged 50 to 59 years were similar [1].

Although the burden of BC has been historically higher among CA compared with other ethnicities, and that AA are known to have the highest burden of early onset BC, the recent changes in mammography and postmenopausal hormone use might have changed the pattern of BC across these two ethnic groups. Furthermore, some risk factors related to reproductive life or anthropometric factors are modifiable and might not have the same prevalence and impact on the different ethnicities and states of the United States. To have a more accurate picture of the evolution of BC burden and to identity potential variations specific to some states or regions within the United States, we have used the database from the Cancer Incidence in Five Continents (CI5) for the period 1998 to 2007 completed by SEER database 2008 to 2010 to assess the

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pattern of BC based on ethnicity, state, and region. In most of the cancer registries included in CI5s, AA and CA are the two ethnicities with the most completed information. Only the states of California and New Mexico have information on Hispanic; however, AA data are not included in this last (New Mexico). For this reason, the present work was restricted to AA and CA.

Methods

Databases and study populations

The study design has been described in details elsewhere [7]. Only cancer registries with available information on both AA and CA were included in the present study. Data on 29 states and the District of Columbia with available information with respect to AA and CA both non-Hispanic were extracted from the volume IX and X of the CI5 which compile data for 5-year periods each; 1998 to 2002 (CI5-IX) and 2003 to 2007 (CI5-X) published in 2007 and 2013, respectively. Figure 1 shows the map of the individual states included in the study grouped by region defined by the US Census Bureau. Of these 29 states and the DC, only 15 states had information available regarding ethnicity from 1998 through 2007.

These data were completed by SEER data for the period 2008 to 2010. However, SEER has 20 registries among which only eight at the state level. Among the registries at the state level, only two are included in the registries covered by the CI5, Louisiana and New Jersey. Therefore, to represent the four regions of our study, the state of New Jersey was used to represent the Northeast, Louisiana for the South, Detroit (Michigan) for the Midwest, and San Francisco and Los Angeles combined for California in the West were used. Greater California was not included in California because it became a SEER registry in 2000, and thus does not contain data for 1998 and 1999. Due to Hurricane Katrina, data from 2005 for Louisiana were excluded from the study. Taking in consideration this last repartition of SEER data, we have at last assessed the pattern of hormonal receptor status over three periods; 1998 to 2002/2000 to 2002, 2003 to 2007, and 2008 to 2010.

Statistical analysis and modeling

Statistical analysis and modeling have being described in our previous work on corpus uteri cancer [7]. Population denominators were derived from the US Census Bureau and showed strong differences when stratified by ethnicity (Fig. S1). Age-standardized rates (ASR) were then adjusted to the world-standard population. To minimizes the tendency of overestimate cancer burden in older age groups with inherently low denominators, and thus provided more accurate calculations of cancer burden distribution across different age groups, we have model the expected number of BC cases in which the age-specific rates were adjusted to the world-standard population [7]. We elected not to use the US standard population for normalization, so that further comparisons with other countries could be performed in future studies. The rate ratio (RR) and 95% confidence interval (CI) were calculated with respect to ethnicity, age group, state, and region. Three age groups were used: younger than 40, 40 to 49, and 50 years or older. All statistical analyses were conducted using Stata.12 software (StataCorp LP, College Station, TX).

Results

ASR and RRs for different study periods (1998–2002 vs. 2003–2007; 2003–2007 vs. 2008–2010)

The present study analyzed 555,403 BC cases for the period 1998 to 2002 (493,721 CA and 61,682 AA); 775,316 cases for the period 2003 to 2007 (686,992 CA and 88,324 AA); and 72,621 cases for the period 2008 to 2010 (60,096 CA and 12,525 AA). The study included 15 states and the District of Columbia for the period 1998 to 2002,

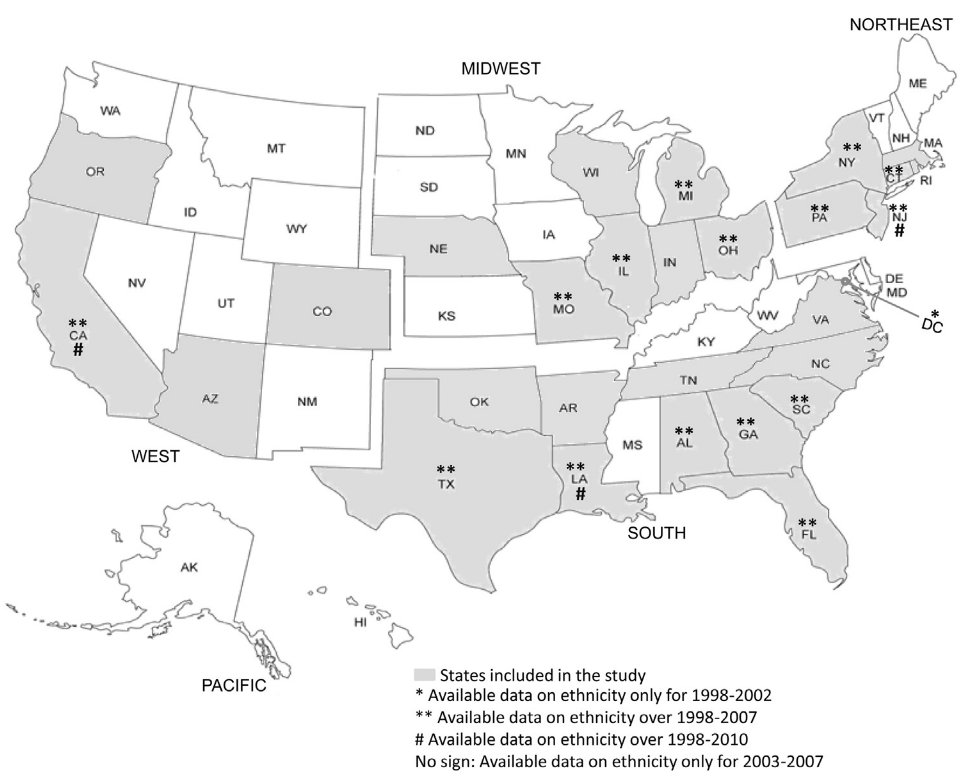

Fig. 1. Geographic distribution of states included in the present study.

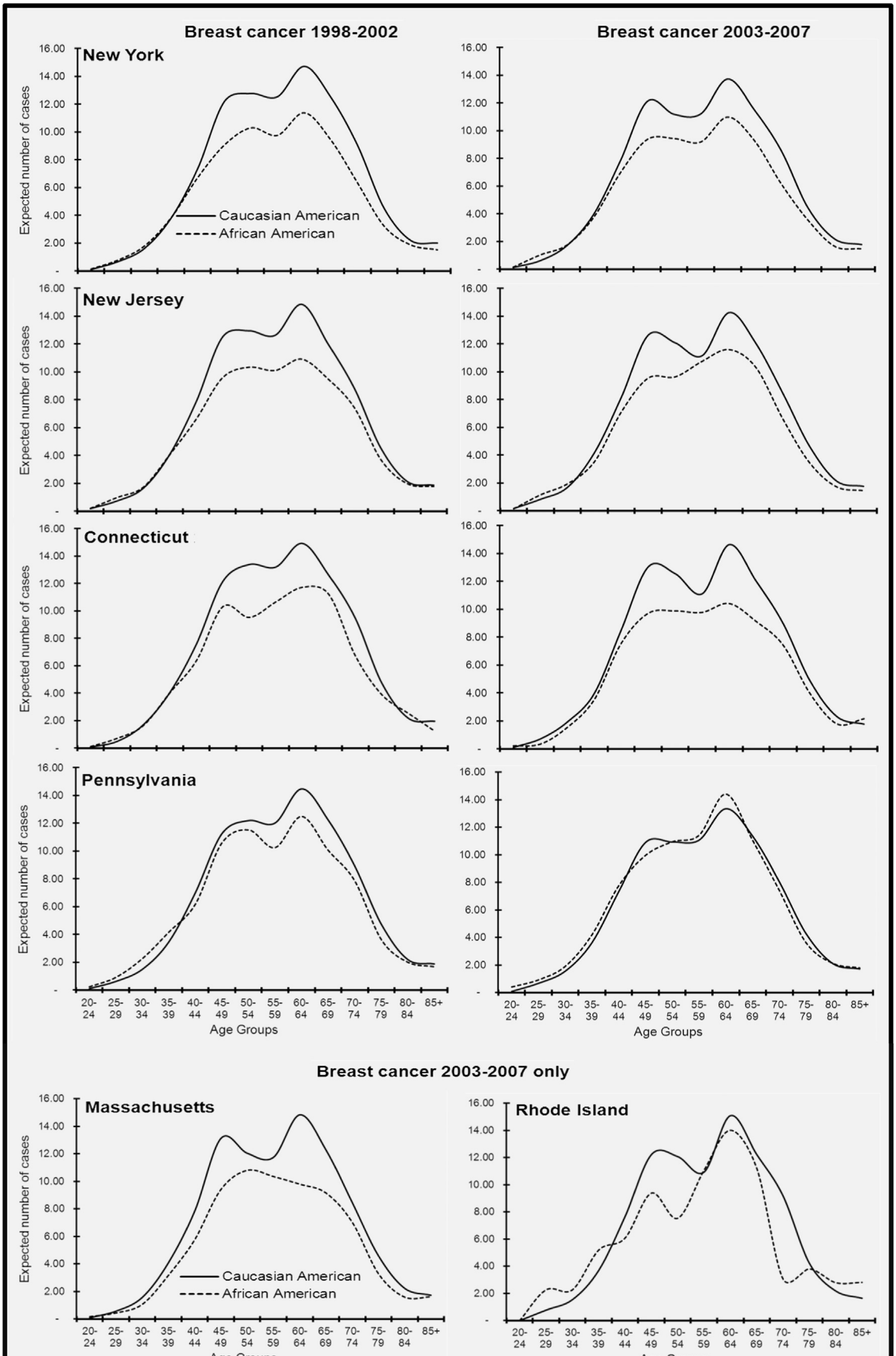

Fig. 2. Distribution of breast cancer burden by ethnicity and age group in Northeastern states. Estimations are based on observed age-specific incidence rates and the world-standard population.

Table 1
2003–2007/1998–2002 rate ratios with respect to age, state, region, and ethnicity

	CA						AA					
	<40	95% CI	40–49	95% CI	≥50	95% CI	<40	95% CI	40–49	95% CI	≥50	95% CI
Northeast												
New Jersey	1.03	0.95–1.11	1.02	0.98–1.07	0.95	0.94–0.97	0.97	0.84–1.12	1.03	0.93–1.15	1.00	0.94–1.06
New York	1.03	0.98–1.09	1.02	0.99–1.05	0.92	0.91–0.93	1.02	0.93–1.12	1.04	0.97–1.11	0.96	0.93–1.00
Connecticut	1.06	0.94–1.19	1.09	1.03–1.16	0.94	0.92–0.97	0.86	0.66–1.12	1.03	0.84–1.27	0.95	0.84–1.06
Pennsylvania	1.05	0.98–1.12	1.00	0.97–1.03	0.92	0.90–0.93	0.98	0.86–1.13	1.06	0.95–1.17	1.06	1.00–1.11
	1.04	1.00–1.07	1.02	1.00–1.04	0.93	0.92–0.94	0.99	0.92–1.05	1.04	0.99–1.09	0.99	0.96–1.01
South												
Alabama	0.98	0.87–1.10	1.01	0.94–1.08	0.94	0.91–0.97	1.05	0.89–1.23	1.18	1.07–1.32	1.07	1.01–1.14
Louisiana	1.04	0.90–1.20	1.03	0.96–1.11	0.94	0.91–0.97	1.07	0.91–1.25	0.96	0.87–1.06	1.05	0.99–1.11
Georgia	1.03	0.94–1.12	1.03	0.98–1.08	0.89	0.87–0.91	1.05	0.94–1.16	1.04	0.97–1.12	1.09	1.04–1.14
Florida	0.99	0.93–1.05	0.93	0.90–0.96	0.88	0.87–0.89	1.10	0.99–1.23	0.94	0.87–1.01	1.00	0.96–1.05
South Carolina	0.99	0.88–1.13	0.98	0.91–1.05	0.93	0.90–0.96	1.01	0.87–1.18	1.11	1.00–1.23	1.09	1.03–1.16
Texas	0.99	0.94–1.04	1.01	0.98–1.04	0.93	0.92–0.94	1.00	0.91–1.10	1.03	0.96–1.11	1.09	1.05–1.14
	1.00	0.96–1.03	0.98	0.97–1.00	0.91	0.90–0.91	1.04	0.99–1.09	1.02	0.98–1.05	1.06	1.04–1.08
Midwest												
Illinois	1.01	0.95–1.08	1.04	1.00–1.08	0.89	0.88–0.91	0.95	0.84–1.07	1.01	0.93–1.10	0.96	0.92–1.01
Michigan	1.09	1.02–1.18	1.03	0.99–1.07	0.88	0.87–0.90	1.16	1.00–1.34	0.98	0.89–1.08	0.99	0.94–1.04
Missouri	1.09	0.98–1.20	1.00	0.95–1.06	0.89	0.87–0.91	0.89	0.73–1.08	0.97	0.84–1.12	0.97	0.90–1.05
Ohio	1.00	0.93–1.07	1.01	0.97–1.05	0.90	0.89–0.92	0.93	0.81–1.07	1.01	0.91–1.12	1.02	0.97–1.08
	1.04	1.00–1.07	1.02	1.00–1.04	0.89	0.88–0.90	0.97	0.90–1.04	1.00	0.95–1.05	0.98	0.96–1.01
West												
California	0.91	0.87–0.95	0.91	0.89–0.93	0.83	0.82–0.84	0.99	0.89–1.09	1.07	0.99–1.15	1.03	0.99–1.08

29 states for the period 2003 to 2007, three states and the city of Detroit for the period 2008 to 2010.

The rates comparison over the two first periods of our study (1998–2002 and 2003–2007) indicates that ASRs for CA decreased in essentially all states. The 2003 to 2007/1998 to 2002 RRs for CA were 0.95 (95% CI, 0.95–0.96) in the Northeast, 0.85 (95% CI, 0.84–0.66) in the West (California), and 0.93 (95% CI, 0.92–0.94) in the South. By contrast, ASRs for AA increased particularly in the South, with 2003 to 2007/1998 to 2002 RR of 1.05 (95% CI, 1.04–1.07). The ASRs for AA were stable in the Northeast and Midwest, with RRs of 1.00 (95% CI, 0.97–1.02) and 0.99 (95% CI, 0.96–1.01), respectively (Table S1). The 2008–2010/2003–2007 RRs for CA indicated little change, with rates of 1.03 (95% CI, 1.00–1.06) in Louisiana, 1.03 (95% CI, 1.01–1.04) in New Jersey, 0.97 (95% CI, 0.94–0.99) in Detroit (Michigan), and 1.00 (95% CI, 0.98–1.01) in California. For AA, the RRs indicated slight increases or essentially stable, with rates of 1.06 (95% CI, 1.01–1.11) in Louisiana, 1.06 (95% CI, 1.01–1.11) in New Jersey, 1.04 (95% CI, 0.99–1.10) in Detroit (Michigan), and 1.00 (95% CI, 0.96–1.04) in California. The median age at diagnosis was quite similar across the 29 states and DC. However, CA were an average of 5 years older than AA at the time of diagnosis, with 62 years for CA and 57 years for AA (P value < .05) (Table S1).

Age-specific BC burden by state according to region

Northeast: New York, New Jersey, Connecticut, Pennsylvania, Massachusetts, and Rhode Island

In all these states, the BC pattern among AA was similar to that of CA, but AA had lower disease burden in all age groups, with the exception of Pennsylvania and Rhode Island, where AA aged younger than 40 years had higher disease burden than that of CA. Among states with available ethnicity information during 1998 to 2002 and 2003 to 2007 (including New York, New Jersey, Connecticut, and Pennsylvania), essentially no change in disease burden patterns were observed for AA or CA (Fig. 2). However, in Pennsylvania, there was an increase in disease burden among AA aged 50 years or older, with RR of 1.06 (95% CI, 1.00–1.11), whereas it decreased among CA aged 50 years or older, with RR of 0.92 (95% CI, 0.90–0.93) (Table 1). For the states with available data over the two first periods, the 2003–2007/1998–2002, RRs were 0.93 (95% CI, 0.92–0.94) for CA of 50 years or older and of 0.99 (95% CI, 0.96–1.01) for AA of the same age range (Table 1). After 2007, the rate of the disease in the same age range was stable for New Jersey (data not shown).

South: Florida, Alabama, Texas, Louisiana, Georgia, South Carolina, North Carolina, Virginia, Arkansas, Tennessee, Delaware, Oklahoma, and the District of Columbia

In all these states with different magnitudes, the well-documented high burden of early onset BC among AA was clearly visible and persistent. A shift in the peak disease burden during the two study periods was observed among AA. In states with available ethnicity information over the 1998 to 2007 (Florida, Alabama, Texas, Louisiana, Georgia, and South Carolina), the highest disease burden during 1998 to 2002 was observed among premenopausal women (<50 years), whereas the highest disease burden during 2003 to 2007 was observed among postmenopausal women (≥50 years) (Fig. 3). However, Alabama was an exception because the AA disease burden increased primarily among premenopausal aged 40 to 49 years, with 2003–2007/1998–2002 RR of 1.18 (95% CI, 1.07–1.32) (Table 1), which resulted in the highest disease burden in premenopausal women during the two study periods. Among states with data only for 2003 to 2007 (North Carolina, Virginia, Arkansas, Tennessee, Delaware, and Oklahoma), the highest disease burden was observed in postmenopausal women. Overall, the 2003–2007/1998–2002 RR was of 0.91 (95% CI, 0.90–0.91) among CA aged 50 years or older and but in contrary to the Northeast, it was of 1.06 (95% CI, 1.04–1.08) for AA of the same age range. After 2007, Louisiana showed an increase in BC rates among AA women aged 50 years or older; the 2008–2010/2003–2007 RR was 1.09 (95% CI, 1.03–1.15).

Midwest: Missouri, Illinois, Michigan, Ohio, Indiana, Wisconsin, and Nebraska

In Missouri and Illinois, AA had a slightly higher burden of early onset BC among premenopausal women for the period 1998 to 2002 compared with that observed for CA. However, the BC pattern among AA was clearly converging with that of CA for the period 2003 to 2007, especially in Missouri and Illinois where the rate of

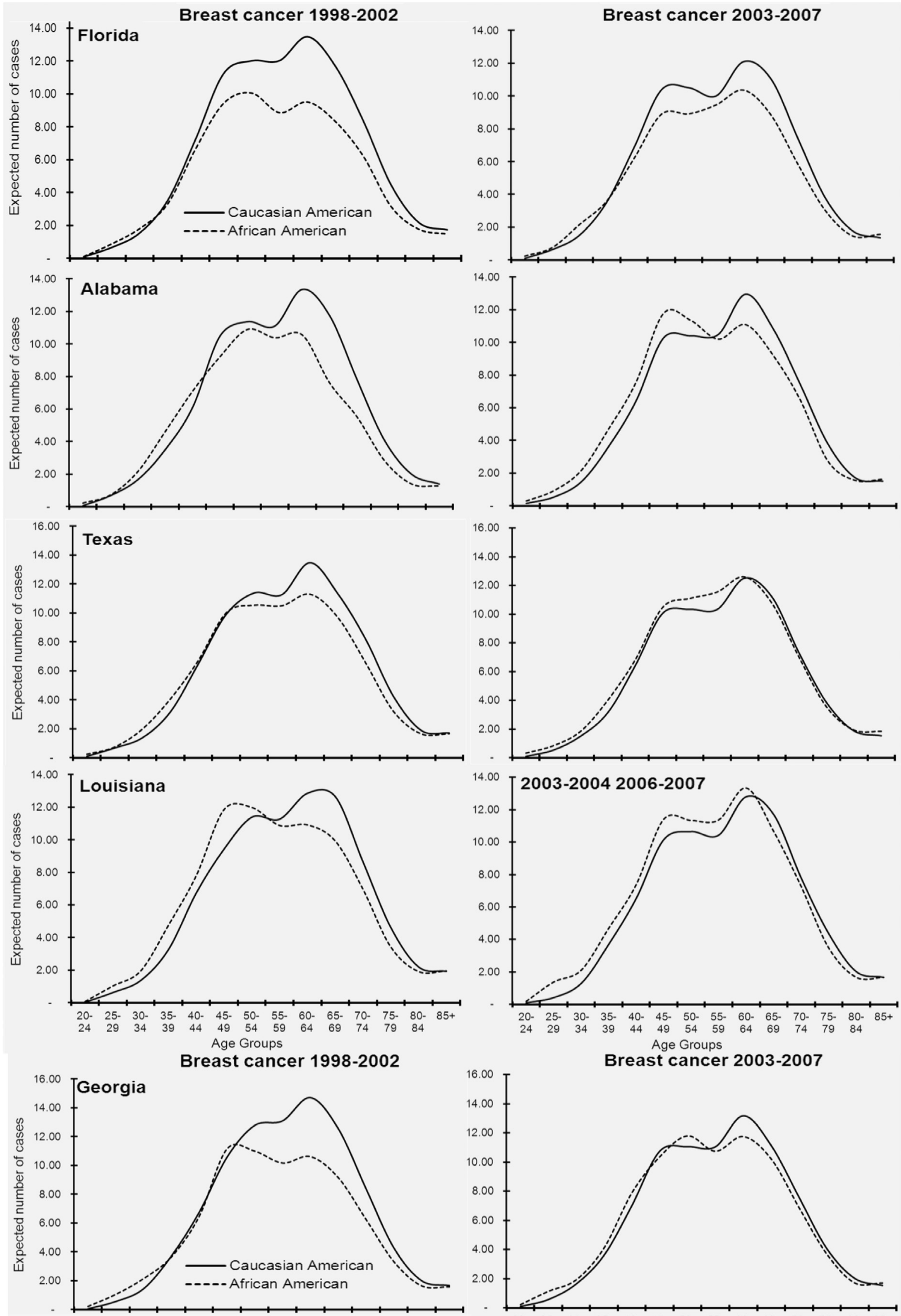

Fig. 3. Distribution of breast cancer burden by ethnicity and age group in Southern states. Estimations are based on observed age-specific incidence rates and the world-standard population.

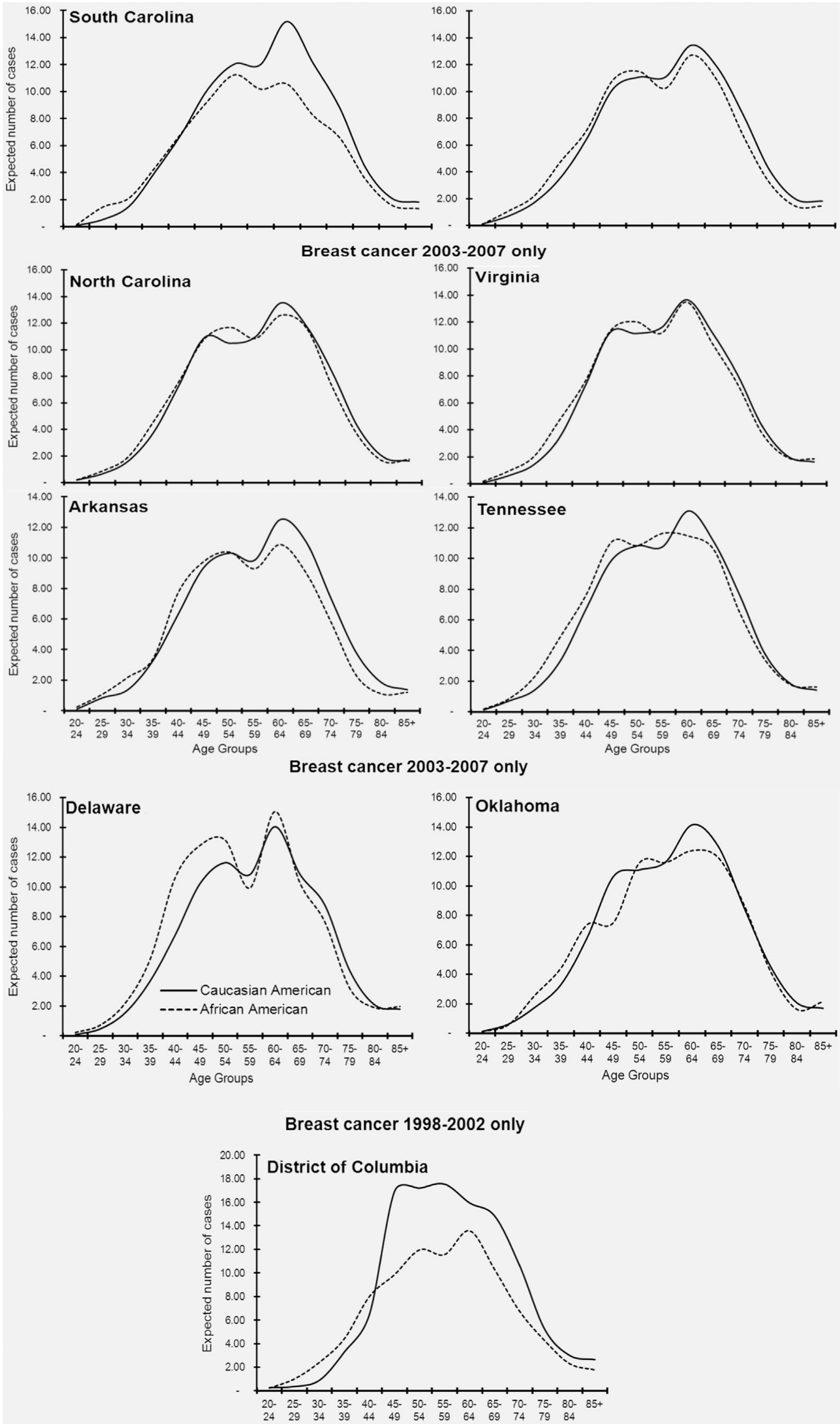

Fig. 3. (*continued*).

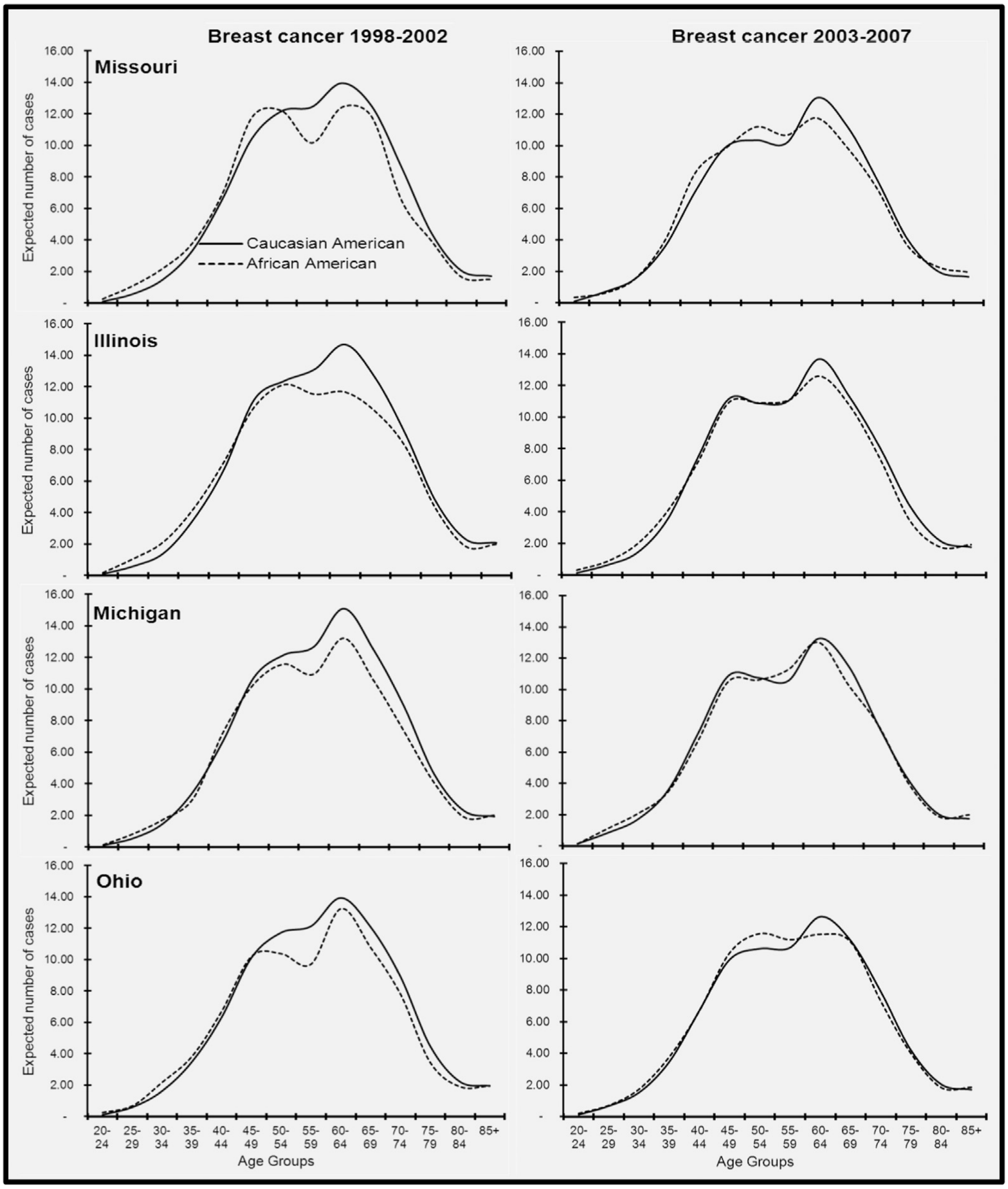

Fig. 4. Distribution of breast cancer burden by ethnicity and age group in Midwestern states. Estimations are based on observed age-specific incidence rates and the world-standard population.

BC were decreasing among CA of 50 years and older 0.89 (95% CI, 0.88–0.91) and 0.89 (95% CI, 0.87–0.91), respectively, whereas for AA of the same age range, the rates were stable; Missouri 0.97 (95% CI, 0.90–1.05), Illinois 0.96 (95% CI, 0.92–1.01) (Fig. 4) (Table 1) (this observation was true for the states of Michigan and Ohio as well). The well-documented early onset BC among AA was observed in Wisconsin with a small magnitude and, the disease pattern was similar for both AA and CA for the period 2003 to 2007. Data from Nebraska were not stable enough to draw any conclusions. Overall, the 2003–2007/1998–2002 RRs were of 0.89 (95% CI, 0.88–0.90) among CA aged 50 years or older and of 0.98 (95% CI, 0.96–1.01) for AA of the same age range. The rate was stable for Detroit (Michigan) after 2007 (data not shown).

West: California, Arizona, Colorado, and Oregon

In California, a very small early onset BC among AA aged younger than 35 years was observed during 1998 to 2002; however, the BC burden among AA was essentially similar to that of CA during 2003 to 2007 (Fig. 5). This was primarily due to decreases in the burden of CA in all age groups (RR of 0.91 [95% CI, 0.87–0.95] for CA aged $<$40 years, and RR 0.83 [95% CI, 0.82–0.84] for CA aged $\geq$50 years), whereas the RRs were stable or slightly increasing among AA (Table 1). Arizona, Oregon, and Colorado had available data only for the period 2003 to 2007. In Arizona, the BC pattern was similar for both AA and CA although AA had a lower disease burden compared with that of CA. In Oregon, AA appear to have a predominance of premenopausal BC compared with that of CA. Data from Colorado were not stable enough to draw any conclusions. As observed in the Northeast, the rate of the disease was stable after 2007.

Age-specific burden for the period 2008 to 2010 SEER data

In New Jersey, the BC patterns among AA and CA were even more convergent than that observed during 2003 to 2007, particularly among those aged 55 years or older (Fig. 6). These trends

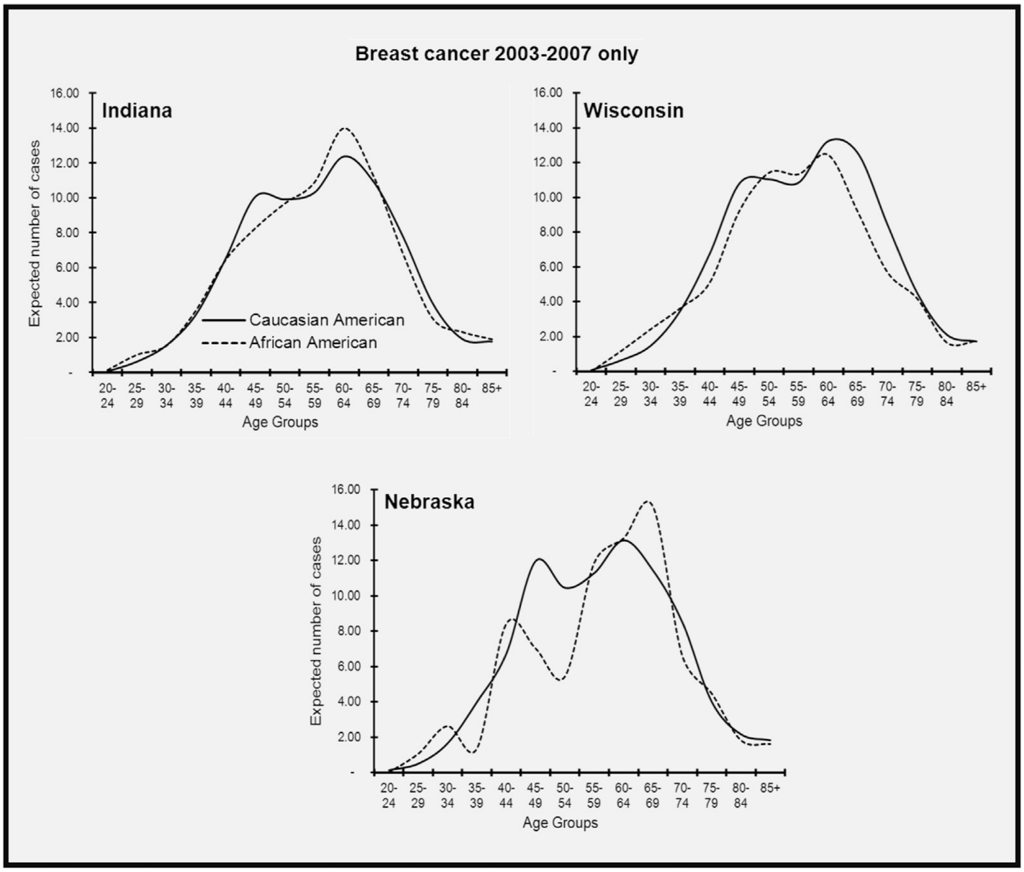

Fig. 4. (*continued*).

continued in Louisiana; the BC patter among AA and CA was similar although AA have a slightly higher burden of the disease in almost all age groups compared with those of CA (Fig. 6). For the Midwest, in Detroit, the BC patterns among AA and CA were essentially the same as that observed in the state of Michigan during 2003 to 2007. For California in the West, the BC burden was quite similar for both ethnicities although the curves slightly differed from those observed for the two last periods.

Persistence of high early onset BC burden among AA in the South and neighboring states

In all states of the South for all study periods, the AA/CA RR was significantly higher than that observed in other regions (with the exception of Florida during 1998–2002). The AA/CA RR was of 1.23 (95% CI, 1.17–1.28) during 1998 to 2002, 1.29 (95% CI, 1.25–1.33) during 2003–2007. The highest discrepancy was observed in Louisiana; 1.52 (95% CI, 1.30–1.78) for the period 2003 to 2007 (Fig. S3). However, for the period 2008 to 2010, the crossover was shifted to the age group 70 to 74 years for Louisiana. In the Northeast, the overall AA/CA RR was 1.10 (95% CI, 1.03–1.16) for the period 1998 to 2002, 0.99 (95% CI, 0.99–1.04) during 2003 to 2007 and 0.99 (95% CI, 0.99–1.04) in New Jersey during 2008 to 2010. In the Midwest, it was of 1.20 (95% CI, 1.12–1.28) during 1998 to 2002, 1.13 (95% CI, 1.08–1.18) during 2003 to 2007 and of 0.99 (95% CI, 0.87–1.14) in Detroit (Michigan) during 2008 to 2010. In the West, the AA/CA RR in California was 1.01 (95% CI, 0.92–1.11) during 1998 to 2002, 1.10 (95% CI, 1.04–1.17) during 2003 to 2007, and 0.99 (95% CI, 0.93–1.08) during 2008 to 2010 (Fig. S3). For the age groups 40 to 49 and 50 years, the AA/CA RR was low or not significantly different in most states and regions for the three study periods (Figs. S4 and S5). In Louisiana during 2008 to 2010, the AA/CA RR was 1.02 (95% CI, 0.94–1.11) for those aged 40–49 years and 0.99 (95% CI = 0.94–1.03) for those aged 50 years or older.

Hormone receptor status by region and ethnicity

The observed convergence was correlating with the pattern of estrogen receptor positive (ER+) or progesterone receptor positive (PR+) over the three periods among CA and AA but not with the pattern of ER negative (ER−) (Fig. S6) or progesterone receptor negative (ER) (data not shown). However, the state of Louisiana and the City of Detroit have shown the highest rates of ER−, PR−, and ERPR− for almost all over the three periods compared with the other regions (Table S2). The AA/CA ER− RR was almost the same for the different states over the two first periods and was around 1.5. But, the strongest AA/CA RR discrepancy with regard to ER− was observed in Louisiana in almost all the three periods and was of 1.73 (95% CI, 1.59–1.89) for the period 2008 to 2010. The same pattern was observed for ERPR−.

Discussion

In this study, we reported that the well-documented early onset BC among AA was mainly restricted to states of the South and some of the Midwest, and was very marginal for the states of the Northeast or the West. Over the different periods, we found a convergence of BC burden among AA and CA. This convergence was mainly due to a decrease of the disease burden among CA aged 50 years or older and an increase of the disease burden among AA of

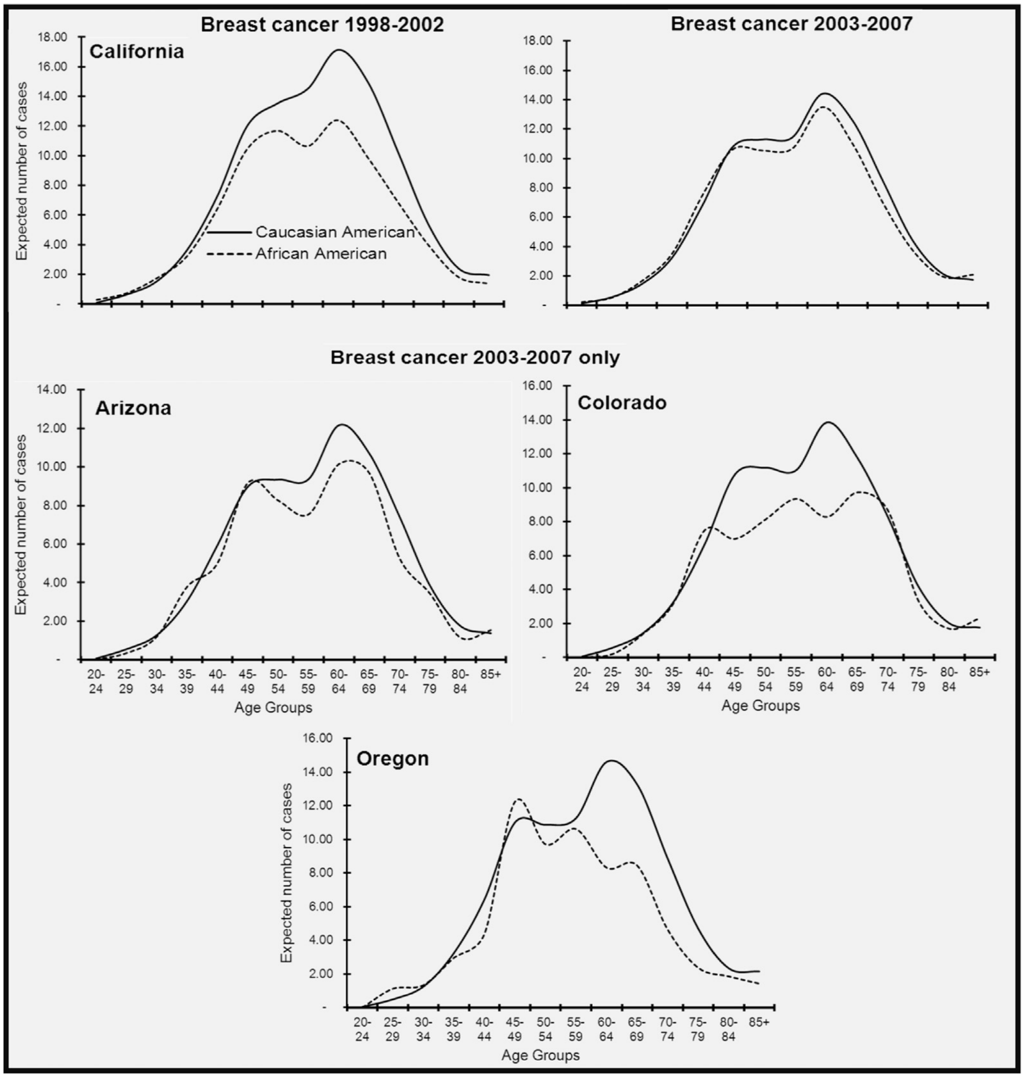

Fig. 5. Distribution of breast cancer burden by ethnicity and age group in Western states. Estimations are based on observed age-specific incidence rates and the world-standard population.

the same age range. However, it should be noted that this increase among AA was stronger in the states of the South. The analyses of the most recent period confirmed the convergence. Also, this pattern correlates with that of estrogen and progesterone receptor positives BC but not with the nonhormonal-dependent BC which was still higher among AA especially AA of the South region where the highest discrepancy in term of the early onset BC and nonhormonal-dependent BC was observed.

It is not known whether the disparity in BC burden among AA and CA is a result of inherited factors. It has been reported that, among women aged 35 years or older, the prevalence of *BRCA1* mutation was highest among AA (16.7%) compared with that of CA (7.2%) or Hispanic (8.9%) [8]. Therefore, it is possible that germ-line mutation of *BRCA* is a key factor in the higher BC burden in younger AA. The different pattern of the disease observed in the other regions compared with the South could be the reflection of race admixture. A report on BC in AA revealed that the average fraction of European admixture in AA was approximately 20% in Eastern states, 25% for participants from West Coast sites, and 19% in the southeast [9]. This correlate with the observation that AA with European ancestry were more likely to develop hormonal-dependent and localized BC [9,10].

Postmenopausal hormone use is a well-established risk factor for BC. After the publication of results from the women health initiative in 2002, which revealed that postmenopausal hormone use was associated to an increase risk of several diseases including BC and that the harm from the use exceed the benefit [11], a drastic decline in the use of postmenopausal hormone has been reported. This decline was stronger among CA who were also known to have the highest prevalence of hormone replacement therapy compared with AA. Thus for CA, the prevalence has drop to about 27% for the period 1999 to 2000, 7% during 2005 to 2006 and 5% during 2009 to 2010 while it was just of 9%, 4%, and 1% for the same periods for AA [12,13]. Therefore, the observed decrease of BC burden among CA aged 50 years or older can be the result of a decline in postmenopausal hormone use. Nevertheless, the use of postmenopausal hormone remains higher among CA [13].

The increased BC burden among older AA can be explained by greater surveillance and detection of the disease. A recent study reported that AA are more likely to undergo mammography screening than CA (64% vs. 59%, respectively) [1]. In the United States, there is an ongoing debate whether mammography screening should begin at the age of 40 or 50 years. The United States Preventive Services Task Force guideline for BC screening

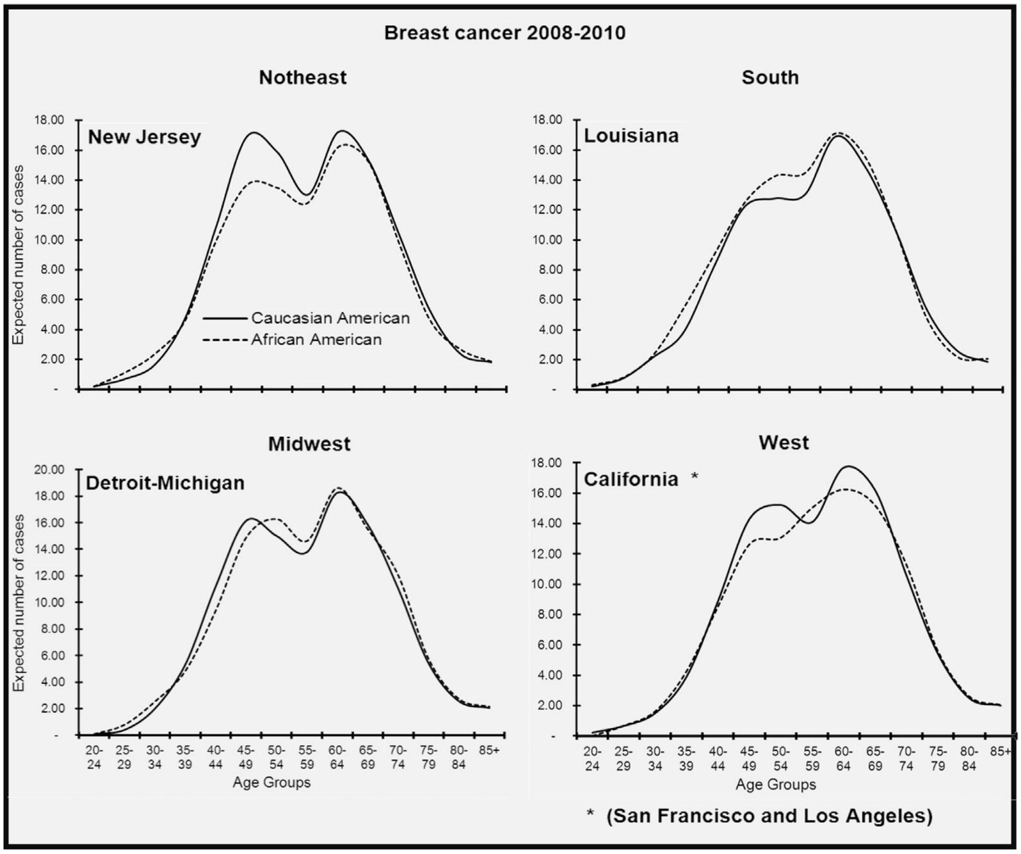

Fig. 6. Distribution of breast cancer burden by ethnicity and age group in states of New Jersey, Louisiana, the city of Detroit and the State of California. Estimations are based on observed age-specific incidence rates and the world-standard population.

(issued in 2009) recommended against routine mammography screening for women aged 40 to 49 years [14]. By contrast, the updated guideline of the American Cancer Society for cancer screening (issued in 2012) recommended annual mammography screening for women aged 40 years or older [15,16]. Our results indicate that early onset BC among AA is persistent in most Southern states for reasons that remain unknown and understudied. Therefore, restricting mammography screening to women aged 50 years or older might result in neglect of high to average-risk women and increase the burden of death. However, the efficiency of mammography screening in saving lives has not being demonstrated [17] and is less accurate for premenopausal women. Therefore, the stability of BC rates observed among women aged younger than 40 years might be due to artificially low cancer detection rates due to technical difficulties in imaging the dense breast tissues of young women [18]. Digital mammography is more effective to identify women at high risk for BC, but there is still no consensus on screening strategy. A recent study demonstrated that a semiannual magnetic resonance imaging was very effective in detecting low-stage BC in genetically defined high-risk women (median age 43 years) (ASCO meeting 2013 http://meetinglibrary.asco.org/content/114172-132). Therefore, a shift toward comprehensive risk assessment should be considered to optimize early detection of the most aggressive forms of BC [19].

The BC patterns we observed were consistent with the reported BC mortality rates with respect to ethnicity and regions [2,20,21]. Based on recent studies of BC statistics from 2006 to 2010 [1], the highest AA/CA RRs for BC mortality in the corresponding regions of our study was observed in South followed by the Midwest; the AA/CA RR mortality was respectively 1.42 and 1.39 compared to 1.15 and 1.18, respectively, in the Northeast and the West. The observed pattern was correlating with that of ER + BC. It has been reported that localized BC was increasing among AA as well as the proportion of ER+ BC [22]. Furthermore, ductal carcinoma *in situ* has been reported to be more likely diagnosed at later age among AA and to be ER+ [23]; therefore, the present increase of the disease burden among AA might not necessarily affect the actual pattern of BC mortality specially the most aggressive BC subtype is still higher among AA specially AA from the South where the mortality from BC is higher among AA.

Although the present study represents one of the most detailed assessments of BC burden at the state level, one of the limitations is the lack of available data on other ethnicities. Furthermore, the relative smaller number of state included in the other regions especially the West region might have bias the picture of BC burden in the region. However, it should be noted that California has the largest and most diverse population of the West.

In summary, this study shows that the patterns of BC burden among AA and CA are converging. These data suggest that in the coming years, BC burden among AA might be higher in all age groups compared with that of CA and point out the need for adapted prevention strategies by states with an emphasis on AA of Southern states. Although an increase in mammography screening as well as the decrease in the use of postmenopausal hormone therapy could have impact on the pattern of BC among AA and CA, respectively; the strongest discrepancy in the early onset BC among AA of the South region remains a mystery. Therefore, to prevent the continuation of a higher mortality among these women, an individual-based approach should also be considered. Indeed, the feasibility of the incorporation of identified high-risk women using Gail model in the construct of a community clinical oncology program has been evaluated [24]. The results showed that the identification of women at high risk for BC can be easily incorporated into an annual screening visit that can allow more appropriate surveillance and prevention at the individual level. At last, detailed assessment of BC cancer pattern with respect to individual states and ethnicity is necessary to gain a better understanding of change in risk-factor distributions.

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Appendix

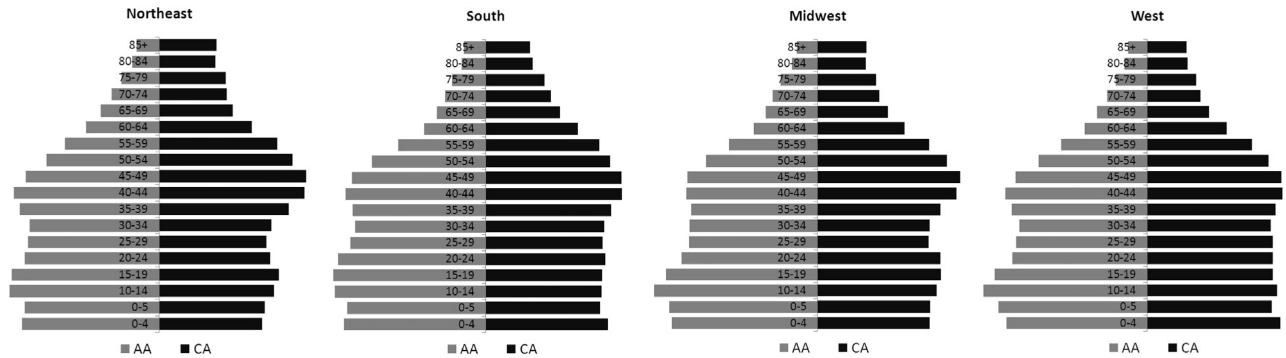


Fig. S1. US population structure with respect to region, age group, and ethnicity, 2003 to 2007.

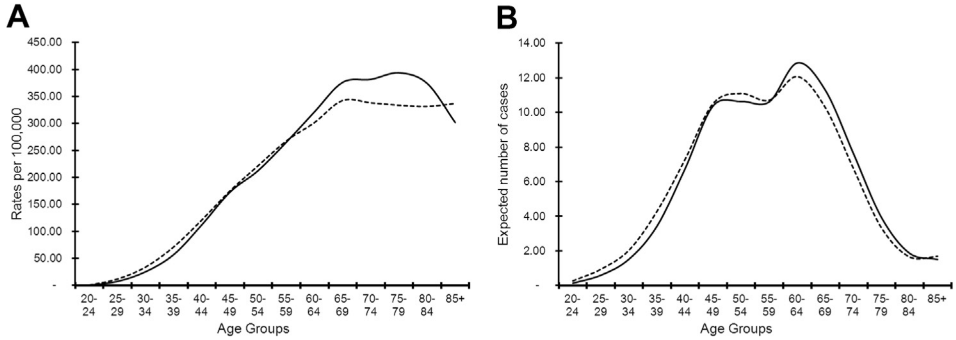
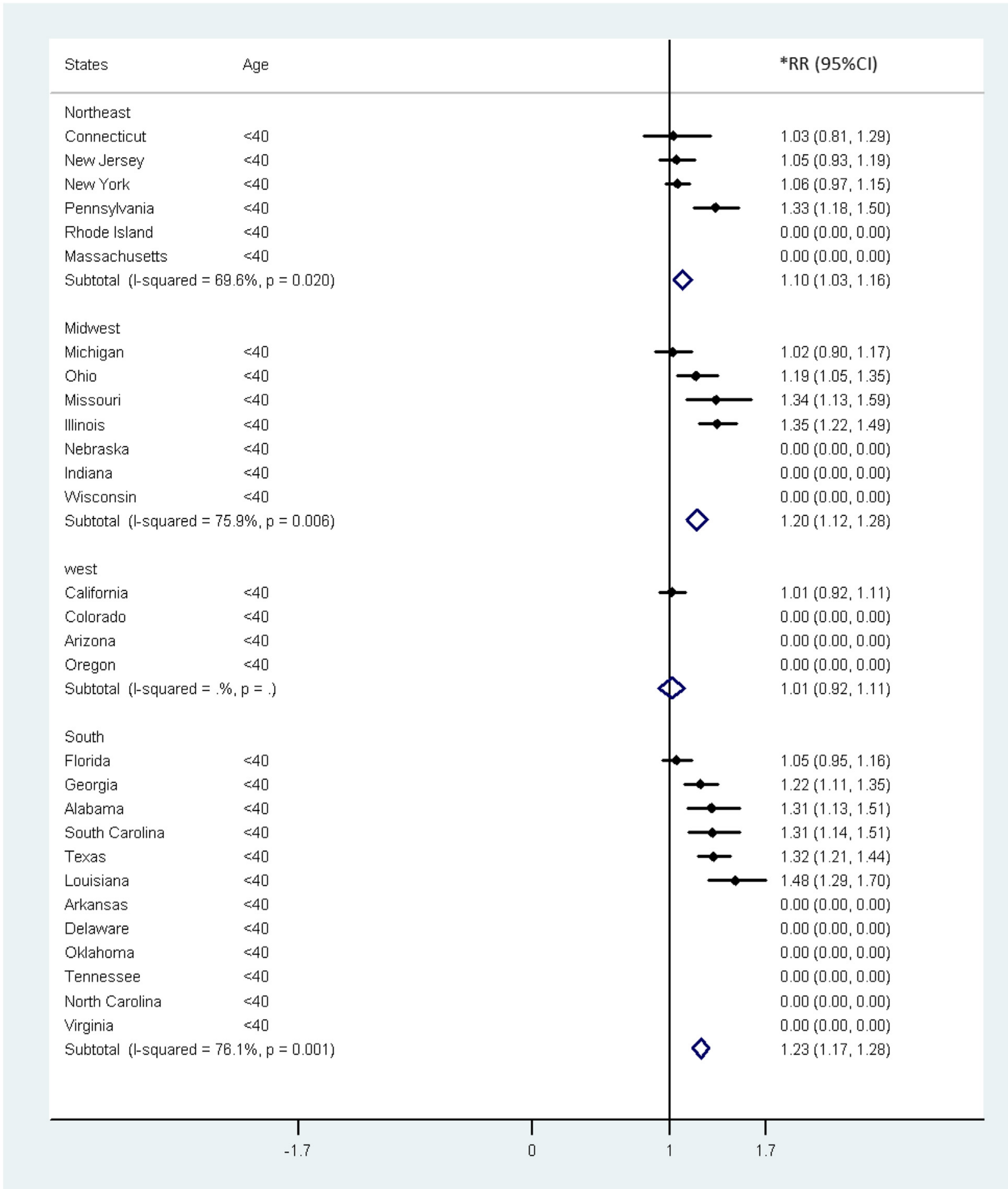


Fig. S2. BC incidence rates with respect to age group and ethnicity in the South region of United States, 2003 to 2007. (A) Age-specific rates per 100,000. (B) BC burden distribution or expected number of cases; estimation based on observed age-specific incidence rates and the world-standard population.

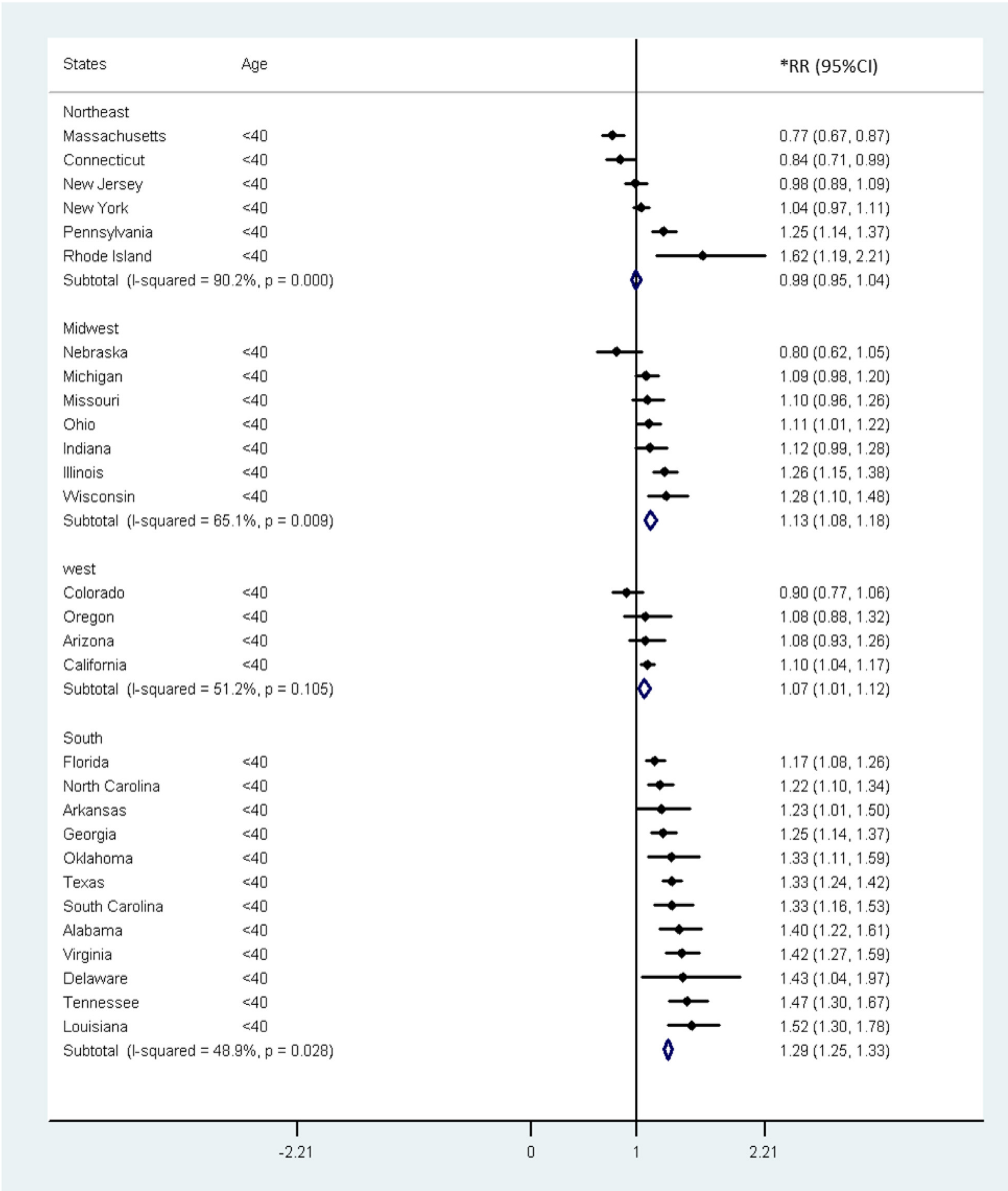
A



***RR :** Rate ratio

Fig. S3. AA/CA BC RRs for women aged younger than 40 years in different states and regions of the United States for the following periods: (A) 1998 to 2002; (B) 2003 to 2007.

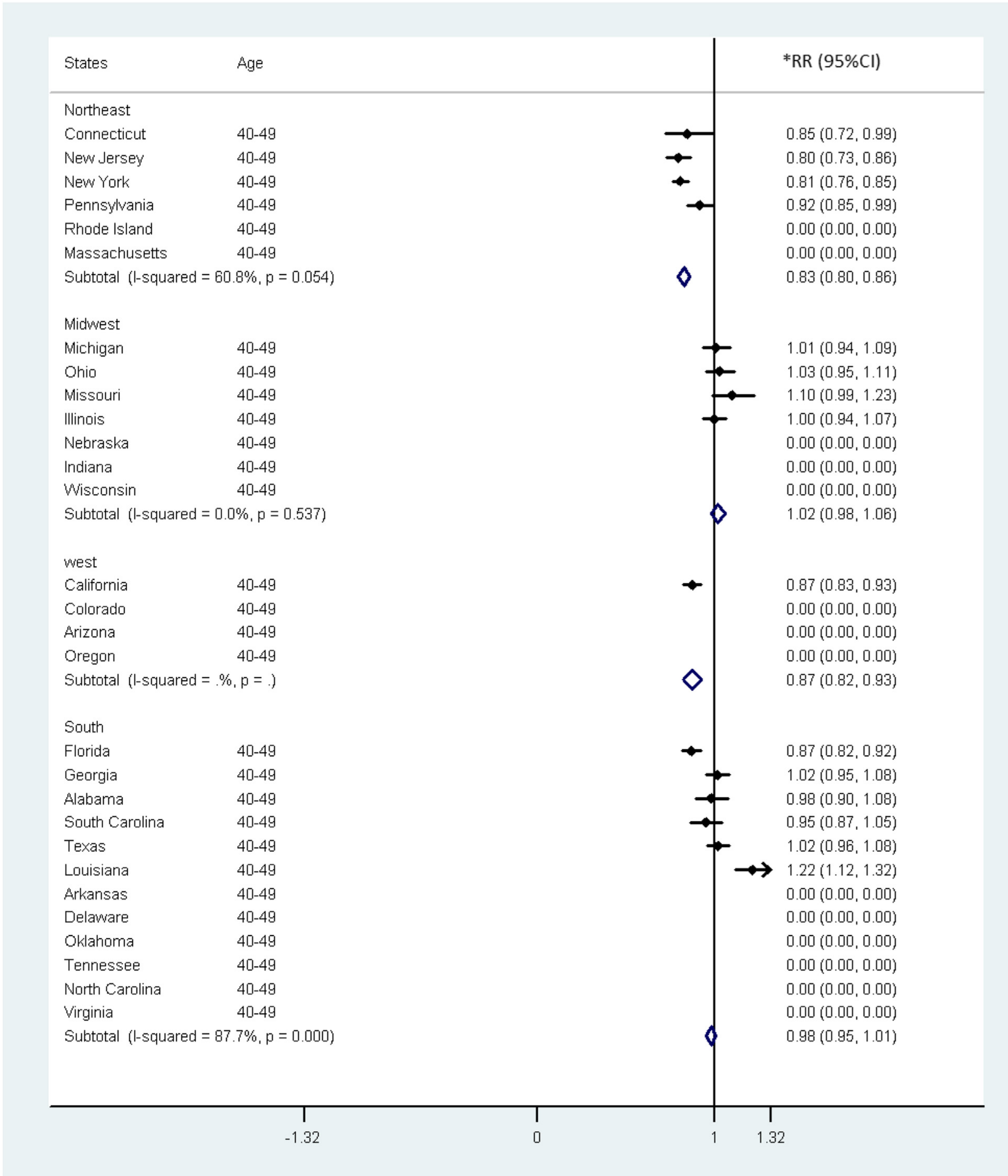
B



***RR :** Rate ratio

Fig. S3. (*continued*).

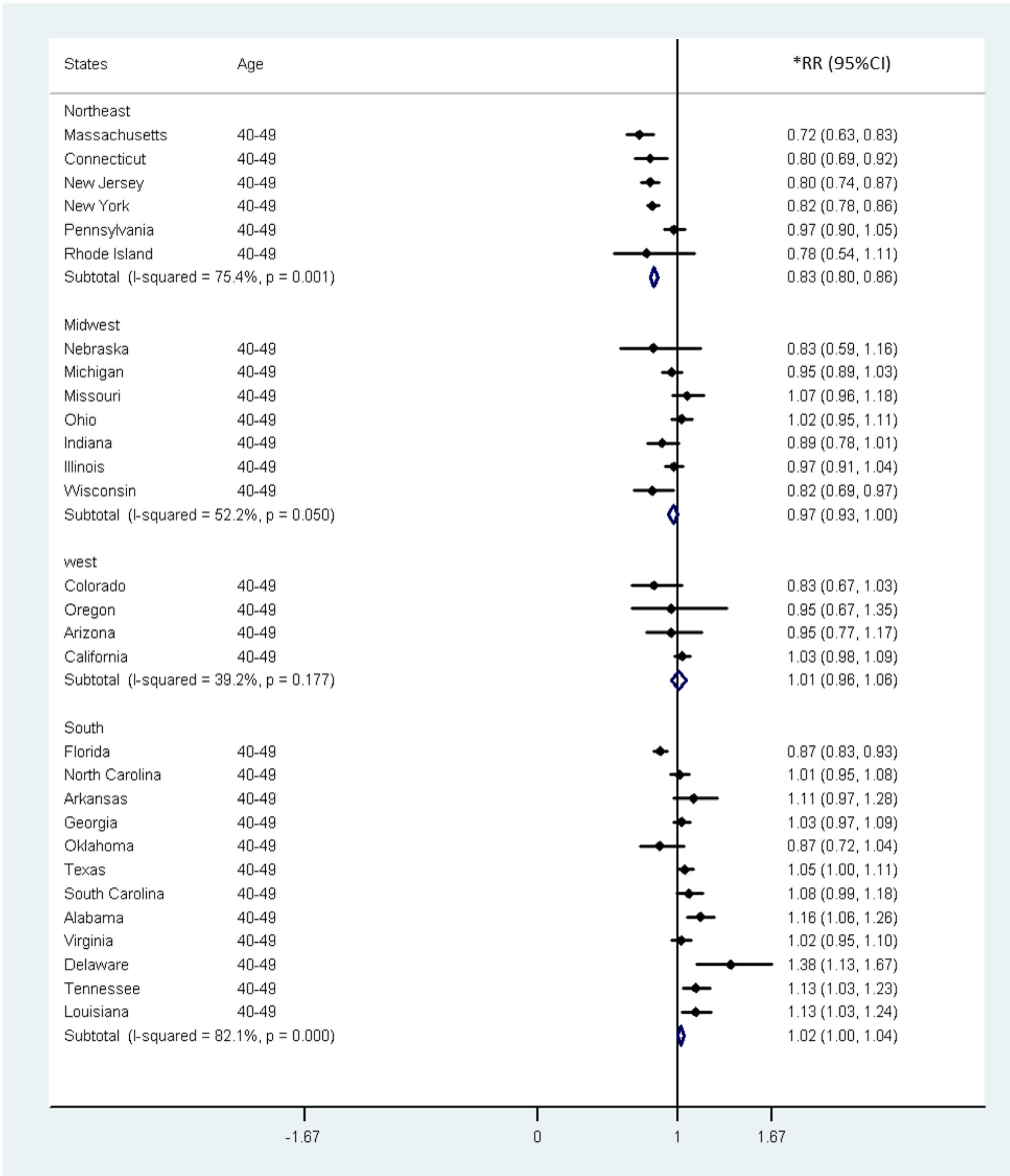
A



***RR :** Rate ratio

Fig. S4. AA/CA BC RRs for women aged 40 to 49 years in different states and regions of the United States for the following periods: (A) 1998 to 2002; (B) 2003 to 2007.

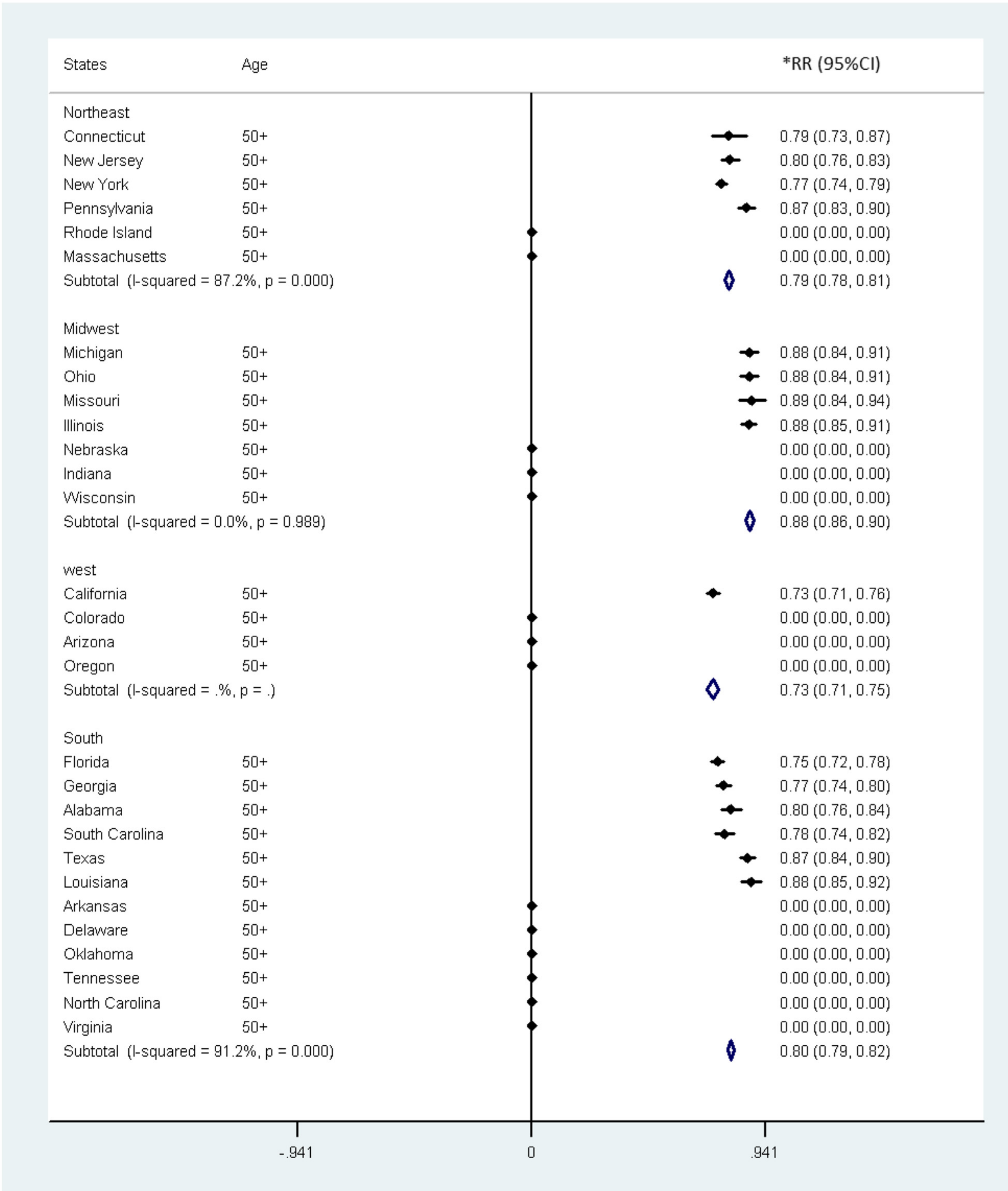
B



***RR :** Rate ratio

Fig. S4. (*continued*).

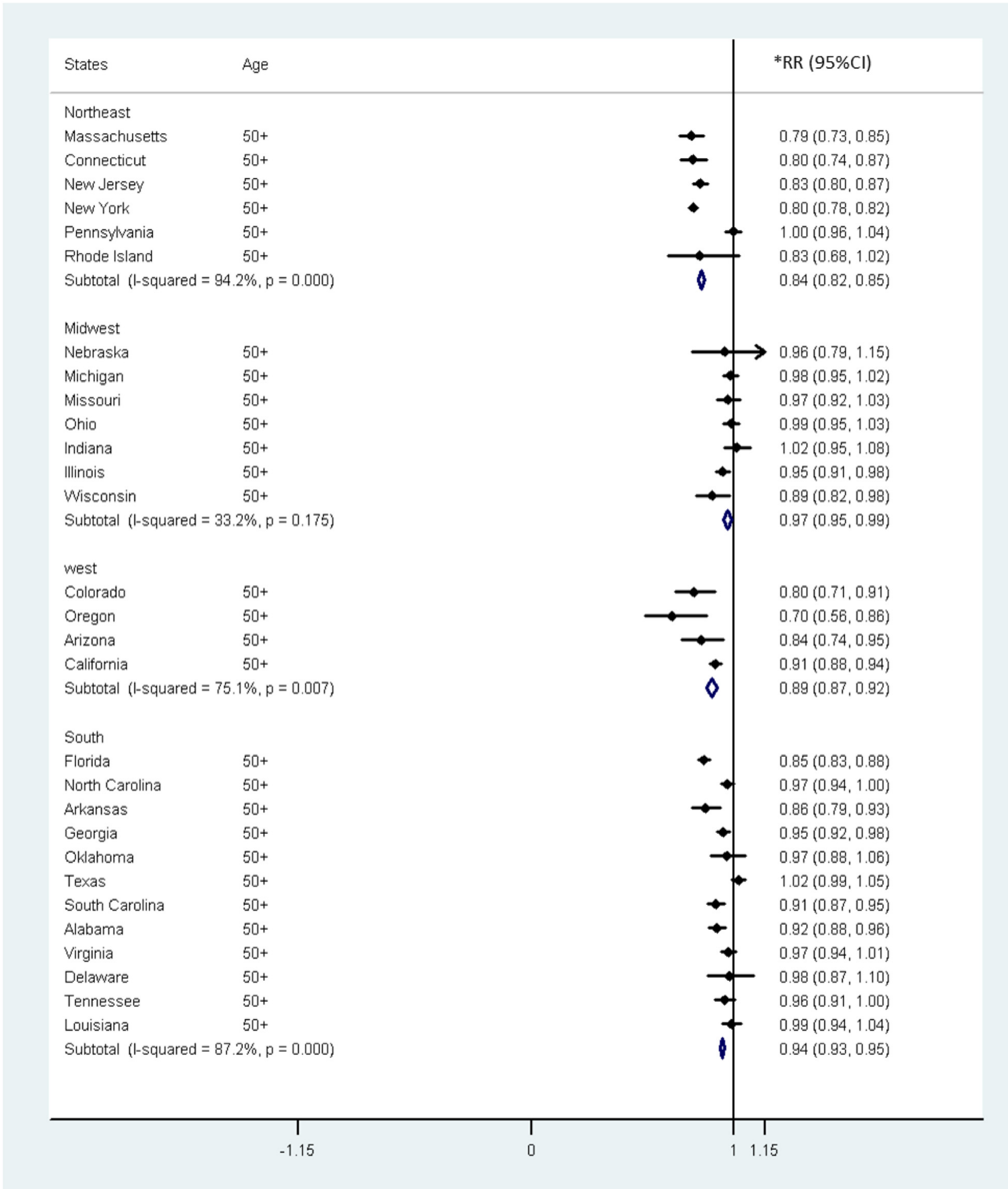
A



***RR :** Rate ratio

Fig. S5. AA/CA BC RRs for women aged 50 years or older in different states and regions of the United States for the following periods: (A) 1998 to 2002; (B) 2003 to 2007.

B



***RR :** Rate ratio

Fig. S5. (*continued*).

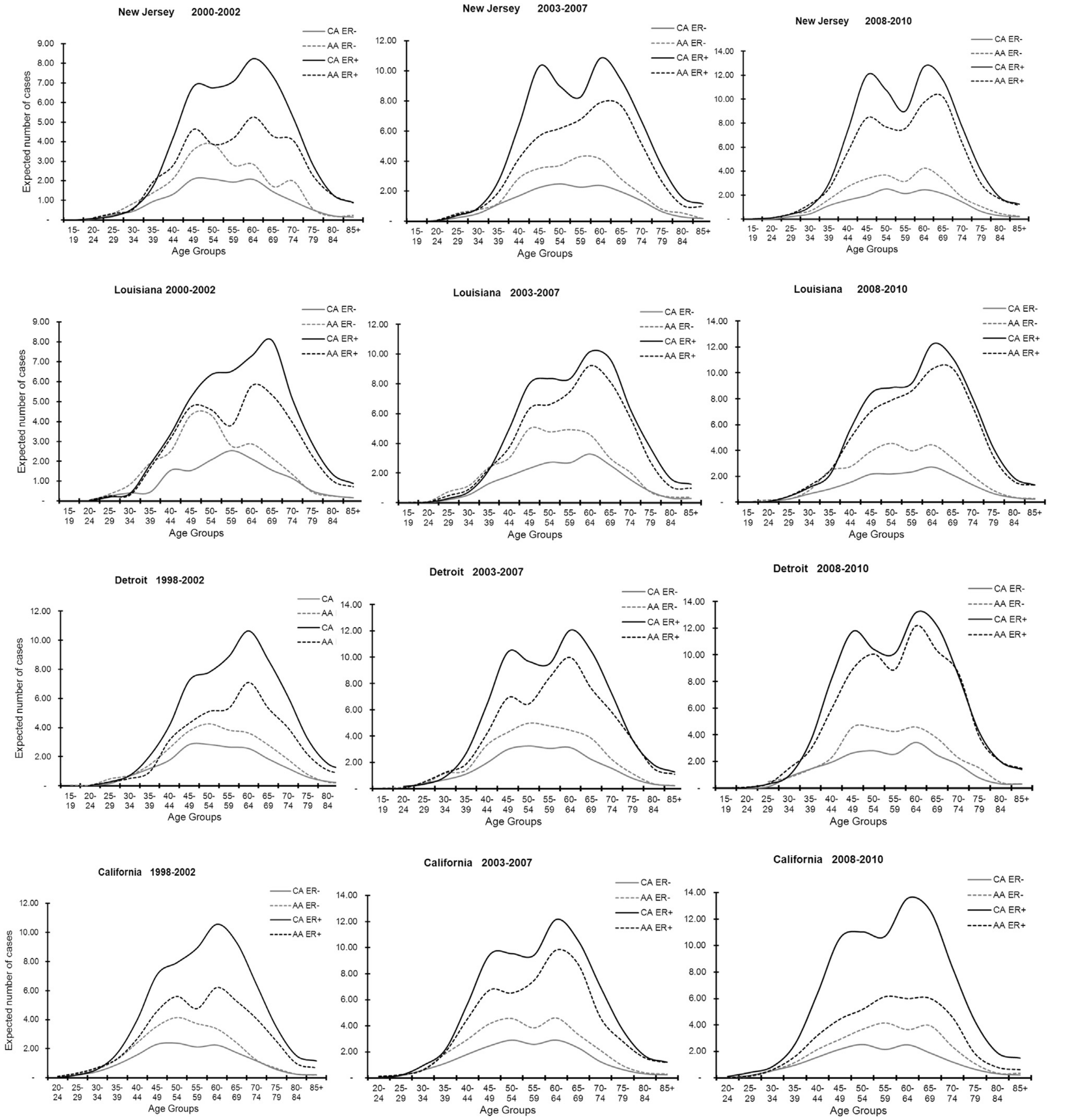


Fig. S6. Distribution of breast cancer burden by ethnicity and age group according the estrogen receptor (ER) status. Estimations are based on observed age-specific incidence rates and the world-standard population.

Table S1
BC incidence rates (age-standardized rate [ASR]), RR, 95% confidence interval (CI), and median age at diagnosis with respect to ethnicity, state, and region in the United States during the periods 1998 to 2002 (′98–′02) and 2003 to 2007 (′03–′07).

Region	State	Number of cases		ASR		RR 95% CI 2003–2007 versus 1998–2002			Median age at diagnosis	
		1998–2002	2003–2007	1998–2002	2003–2007	RR	95% CI		1998–2002	2003–2007
CA										
Northeast	New York	58,020	55,852	95.98	90.90	0.95	0.94	0.96	63	62
	New Jersey	26,794	26,873	96.76	94.19	0.97	0.96	0.99	62	62
	Connecticut	12,386	12,502	98.57	96.51	0.98	0.96	1.00	63	62
	Pennsylvania	44,441	42,517	92.61	87.05	0.94	0.93	0.95	66	63
	Massachusetts	—	23,046	—	95.24	—	—	—	—	62
	Rhode Island	—	3847	—	93.41	—	—	—	—	62
South	Florida	56,396	53,982	90.20	80.93	0.90	0.89	0.91	66	64
	Alabama	11,396	11,419	85.31	81.54	0.96	0.93	0.98	63	62
	Texas	49,755	52,461	85.24	80.71	0.95	0.94	0.96	61	61
	Louisiana	10,202	7992	87.01	83.63	0.96	0.93	0.99	64	63
	Georgia	18,594	19,356	92.54	85.34	0.92	0.90	0.94	61	61
	South Carolina	10,251	10,828	91.57	86.29	0.94	0.92	0.97	62	62
	North Carolina	—	23,103	—	87.29	—	—	—	—	62
	Virginia	—	19,594	—	87.56	—	—	—	—	61
	Arkansas	—	7641	—	79.44	—	—	—	—	63
	Tennessee	—	16,684	—	82.60	—	—	—	—	62
	Delaware	—	2419	—	87.18	—	—	—	—	63
	Oklahoma	—	10,717	—	89.27	—	—	—	—	63
	District of Columbia	754	—	115.20	—	—	—	—	59	—
Midwest	Missouri	17,697	17,292	90.31	83.23	0.92	0.90	0.94	63	62
	Illinois	36,402	34,809	94.37	87.45	0.93	0.91	0.94	63	62
	Michigan	29,961	28,985	93.24	86.02	0.92	0.91	0.94	63	62
	Ohio	36,041	34,761	89.67	83.42	0.93	0.92	0.94	63	62
	Indiana	—	18,005	—	81.04	—	—	—	—	62
	Wisconsin	—	18,020	—	87.66	—	—	—	—	63
	Nebraska	—	5671	—	87.81	—	—	—	—	63
West	California	74,631	87,622	105.15	89.40	0.85	0.84	0.86	63	61
	Arizona	—	15,264	—	75.78	—	—	—	—	63
	Colorado	—	13,181	—	86.84	—	—	—	—	59
	Oregon	—	12,549	—	90.52	—	—	—	—	63
AA										
Northeast	New York	7618	9103	76.09	74.68	0.98	0.95	1.01	58	58
	New Jersey	3104	3359	78.87	78.95	1.00	0.95	1.05	57	58
	Connecticut	791	875	80.76	77.44	0.96	0.87	1.06	57	57
	Pennsylvania	3566	3984	83.82	87.96	1.05	1.00	1.10	60	59
	Massachusetts	—	962	—	73.70	—	—	—	—	57
	Rhode Island	—	144	—	81.45	—	—	—	—	57
South	Florida	4975	5930	71.53	71.26	1.00	0.96	1.03	56	57
	Alabama	2772	3316	74.47	81.62	1.10	1.04	1.15	57	56
	Texas	5499	6887	78.90	84.66	1.07	1.04	1.11	56	56
	Louisiana	3757	3159	85.44	87.97	1.03	0.98	1.08	57	57
	Georgia	5235	6910	78.37	84.15	1.07	1.04	1.11	54	55
	South Carolina	2974	3541	77.49	84.10	1.09	1.03	1.14	57	57
	North Carolina	—	5587	—	86.90	—	—	—	—	57
	Virginia	—	4504	—	88.75	—	—	—	—	57
	Arkansas	—	994	—	74.45	—	—	—	—	56
	Tennessee	—	2716	—	84.73	—	—	—	—	56
	Delaware	—	527	—	94.84	—	—	—	—	55
	Oklahoma	—	705	—	86.73	—	—	—	—	58
	District of Columbia	1299	—	88.35	—	—	—	—	62	—
Midwest	Missouri	1685	1836	86.15	83.23	0.97	0.90	1.03	57	58
	Illinois	5240	5435	87.51	84.99	0.97	0.94	1.01	58	58
	Michigan	3770	4064	84.94	84.75	1.00	0.95	1.04	58	58
	Ohio	3548	3925	83.00	83.83	1.01	0.97	1.06	60	58
	Indiana	—	1384	—	80.82	—	—	—	—	59
	Wisconsin	—	672	—	79.18	—	—	—	—	56
	Nebraska	—	157	—	80.63	—	—	—	—	59
West	California	5849	6764	81.35	84.46	1.04	1.00	1.08	57	58
	Arizona	—	381	—	66.48	—	—	—	—	56
	Colorado	—	377	—	70.77	—	—	—	—	57
	Oregon	—	126	—	69.40	—	—	—	—	55

Table S2
Age-standardized rate of ER and PR status by period, state, and ethnicity

	2000–2002				2003–2007				2008–2010			
	ER+		ER−		ER+		ER−		ER+		ER−	
	CA	AA	CA	AA	CA ER+	AA ER+	CA ER−	AA ER−	CA ER+	AA ER+	CA ER−	AA ER−
New Jersey	53.29	36.45	14.52	22.56	71.27	51.45	17.41	27.13	82.80	65.67	17.09	26.84
Louisiana	49.52	37.63	14.61	24.30	65.65	55.73	20.33	33.22	74.47	66.93	17.47	30.28
Detroit	62.27	39.99	18.74	26.63	76.51	59.31	21.69	32.63	86.62	76.99	21.28	31.12
California	62.86	39.76	16.10	24.09	73.74	56.91	19.35	30.05	84.83	41.19	16.90	26.42
	PR+		PR−		PR+		PR−		PR+		PR−	
New Jersey	43.44	28.91	24.36	30.10	60.93	43.26	27.75	35.32	72.77	56.09	27.12	36.43
Louisiana	40.88	31.16	23.26	30.76	55.89	45.29	30.09	43.66	64.50	58.37	27.44	38.85
Detroit	52.15	32.71	28.86	33.90	64.95	46.98	33.25	44.96	74.89	67.12	33.02	41.00
California	54.22	46.88	24.73	40.08	62.79	46.88	30.30	40.08	71.92	44.31	29.81	40.31
	ERPR++		ERPR−−		ERPR++		ERPR−−		ERPR++		ERPR−−	
New Jersey	42.28	27.15	13.35	20.81	60.02	41.79	16.50	25.66	71.80	54.74	16.12	25.50
Louisiana	39.74	28.80	13.48	21.93	54.51	43.32	18.95	31.25	63.15	55.99	16.11	27.90
Detroit	50.61	31.12	17.20	25.04	63.25	45.25	20.00	30.90	73.86	64.25	20.26	28.26
California	52.55	31.90	14.42	21.74	61.78	45.48	18.34	28.64	71.21	34.93	16.19	25.54